

T24-03-10322
BSC-CS2

THE UNIVERSITY OF DODOMA
COLLEGE OF INFORMATICS AND VIRTUAL EDUCATION
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CP 221: INTERNET PROGRAMMING AND APPLICATIONS I
TEST ONE (50 Marks) **TIME: 60 Minutes**

Question One (25 Marks)

A university is developing an online learning system that presents lesson materials as an interactive slideshow. Each slide contains images stored in the images folder. The system must be developed using HTML for the page structure, CSS for styling, and JavaScript for interactivity. The JavaScript implementation should be dynamic so that the slideshow can support any number of slides without requiring changes to the JavaScript code whenever new slides are added.

The slideshow must appear inside a centred container with a width of 700 pixels and a height of 400 pixels. The container should include a visible border, rounded corners, a light background colour, and a shadow effect. Only one slide should be visible at a time. A page title should be displayed at the top of the page, while a status area below the slideshow should display the current slide number and the total number of slides in the format "Slide 1 of 5." Below the slideshow, there should be four control buttons labelled Previous, Next, Play, and Pause. Another row of numbered buttons should allow users to jump directly to a specific slide. CSS should be used to style all buttons consistently and to highlight the currently active numbered button.

When the page loads, the first slide should be displayed automatically. Clicking the Next button should display the next slide, and if the current slide is the last one, the slideshow should continue from the first slide. Clicking the Previous button should display the previous slide, and if the current slide is the first one, the slideshow should continue from the last slide. Clicking a numbered button should immediately display the corresponding slide.

The Play button should start an automatic slideshow that changes slides every five seconds, while the Pause button should stop the automatic slideshow. If a slide containing a video becomes hidden, the video should pause automatically. In addition, users should be able to navigate through the slideshow using the left and right arrow keys on the keyboard. The following directory structure is provided for the project: slider/index.html, presentation.js, presentation.css, and the images folder.

```
<!DOCTYPE html>
<html>
<head>
  <title>Interactive Web Technologies Lesson</title>
  <link rel="stylesheet" type="text/css" href="presentation.css" />
  <script type="text/javascript" src="presentation.js"></script>
</head>
<body onload="initializePresentation()">

  <h1>Interactive Web Technologies Lesson</h1>
  <div id="presentation">
    <div class="slide" style="display:block;">Slide 1</div>
    <div class="slide" style="display:none;">Slide 2</div>
    <div class="slide" style="display:none;">Slide 3</div>
    <div class="slide" style="display:none;">Slide 4</div>
    <div class="slide" style="display:none;">Slide 5</div>
  </div>
```



```

<div id="controls">
  <button onclick="changeSlide(-1)">Previous</button>
  <button onclick="changeSlide(1)">Next</button>
  <button onclick="startAutoPlay()">Play</button>
  <button onclick="stopAutoPlay()">Pause</button>
</div>
<div id="indicators">
  <button onclick="goToSlide(0)">1</button>
  <button onclick="goToSlide(1)">2</button>
  <button onclick="goToSlide(2)">3</button>
  <button onclick="goToSlide(3)">4</button>
  <button onclick="goToSlide(4)">5</button>
</div>
<div id="status"></div>
</body>
</html>

```

Write the complete *presentation.css* file to style the slideshow and navigation controls as described. Then write the complete *presentation.js* file to implement all required functionality, including *slide navigation*, *wrap-around behaviour*, *autoplay*, *pause*, *keyboard navigation*, *status updates*, and *active button highlighting*. Your solution must work correctly even if more slides are added later to the presentation container.

Question Two (25 Marks)

To modernize and automate the seat booking process for the Tanzania Railways Corporation Standard Gauge Railway (SGR), you are required to design and develop a responsive web-based seat booking system. The system should enable passengers to reserve train seats online by entering their personal information, including first name, middle name, surname, age, gender, National Identification Number, email address, and phone number. The system must also provide passengers with the option to select their preferred train carriage category from the following options: Economy Class Tsh 40,000, Business Class Tsh 80,000 and Royal Class Tsh 120,000.

Using JavaScript, the application should dynamically update and display the selected fare and total booking cost immediately after a carriage category is chosen, ensuring that passengers can clearly view the amount payable before confirming their reservation. Upon successful submission of the booking form, the system should display a confirmation alert message notifying the passenger that the booking has been completed successfully.

In addition, the system must include appropriate CSS styling requirements as follows:

- All text within the body of the webpage should appear in blue color and be displayed at twice the default font size.
- The page header should use an image named *background.png* as its background. The image must cover the entire header container and begin positioning from the top-left corner.
- The header section should prominently display a bolded title labeled "Tanzania Railway Company".
- A footer section should be included with a black background and white-colored copyright text reading: "@2026, All Rights Reserved."
- The submit button must include hover effects for improved user interaction.
- The webpage design should be fully responsive by adjusting the form layout appropriately for smaller screen devices such as tablets and smartphones.